

Voice-to-Order AI System – through AI-based speech understanding

The project should be developed with a modern user interface and a robust backend. It should also include a comprehensive [README](#) file containing all essential project information such as setup instructions, features, usage guidelines, API details, and technical implementation notes.

Introduction

Purpose

This document defines the software requirements for a **Voice-to-Order AI System**.

The purpose of the system is to allow users to place food orders through voice commands, process spoken input using Artificial Intelligence, detect the user's intent, extract order details, and generate structured restaurant orders through a smart application interface the order item will be put into cart and place the order .

Scope

The Voice-to-Order AI System will provide intelligent ordering functionality including:

- Any language of Voice input capture
- Speech-to-text conversion
- Intent detection
- Entity extraction
- Order creation from spoken commands
- Support for mixed-language commands
- Restaurant and previous-order context handling
- Order will be seen on the add to cart
- Place the order

The system is designed for convenience, automation, and scalability.

Overall Description

Product Perspective

The application is a standalone AI-based system that processes voice commands through APIs or a user interface.

It can later be integrated with:

- Restaurant ordering platforms
- User authentication systems
- Delivery applications
- CRM or customer profile systems

Product Functions

The system shall:

- Capture voice input from the user
- Convert speech to text
- Detect user intent such as ordering, reordering, or modifying an order
- Extract entities such as item name, quantity, restaurant name, and special instructions
- Support mixed-language input such as “2 dosa order karna hai”
- Use previous order history when the user says “from my last restaurant”
- Generate a structured order request
- Return order confirmation and parsed details

Operating Environment

- Server-side application
- Runs on standard operating systems such as Windows, Linux, and macOS
- Uses a database for storing user history and orders
- Uses AI/ML libraries and speech models for voice understanding
- Supports web-based or mobile-based front end

Design and Implementation Constraints

- The system must use a backend framework
- Data must be stored persistently
- Responses must be in JSON format
- The system must support speech-to-text processing
- AI/ML models must be used for intent detection and entity extraction
- The system must handle mixed-language and colloquial commands

Functional Requirements

Voice Input Capture

Input:

- Audio input from microphone and sent to the model via the api

The system shall allow users to provide voice commands.

Output:

- Audio received confirmation
- Processed voice session ID

Speech-to-Text Conversion

The system shall convert spoken audio into text.

Input:

- Voice command such as:
 - “Order 2 paneer pizzas from my last restaurant”
 - “2 dosa order karna hai”

Output:

- Transcribed text
- Confidence score

Intent Detection

The system shall detect the user’s intent from the transcribed text.

Possible intents:

- Place order
- Reorder previous item
- Modify order
- Cancel order
- Ask menu details

Output:

- Detected intent
- Confidence score

Entity Extraction

The system shall extract important order details from the user command.

Extracted entities may include:

- Item name
- Quantity
- Restaurant reference
- Size
- Add-ons
- Special instructions
- Language indicators

Output:

- Structured order data

Context Understanding

The system shall use past order history and restaurant context when needed.

Example:

- “Order 2 paneer pizzas from my last restaurant”

The system shall identify:

- Quantity = 2
- Item = paneer pizza
- Restaurant = last ordered restaurant

Mixed-Language Support

The system shall understand mixed-language or Hinglish input.

Example:

- “2 dosa order karna hai”

The system shall identify:

- Quantity = 2
- Item = dosa
- Intent = place order

Order Confirmation

The system shall show the parsed order before final submission.

Output:

- Confirmed item list
- Total quantity
- Restaurant name
- Order summary

View Order History

The system shall allow retrieval of previous orders.

Output:

- Order ID

- Restaurant name
- Items ordered
- Date and time
- Status

Input Validation

The system shall validate all required fields before processing.

- Audio file must not be empty
- Speech must be understandable enough for processing
- Required order details must be present
- Restaurant or user context must be available when needed

Error Handling

The system shall return appropriate error messages when:

- Audio upload fails
- Speech-to-text fails
- Intent detection fails
- Entity extraction fails
- Restaurant or order history is not found
- Internal errors occur

Non-Functional Requirements

Performance Requirements

- The system shall process voice commands within acceptable time limits under normal load
- Speech recognition and intent extraction should be near real-time

Reliability

- The system shall ensure data consistency
- The system shall not lose order history during normal operation

Usability

- UI responses shall be clear and human-readable
- Order summaries shall be easy to understand
- Errors shall be descriptive and meaningful

Security

- The system shall prevent invalid or malformed data from being stored
- Audio and user data shall be handled securely
- Authentication is not required in the current version, but may be added later

Maintainability

- The system shall be modular
- The system shall allow future enhancement without major restructuring
- Speech models and NLP logic should be replaceable or upgradable

Data Requirements

User Entity

Field	Description
ID	Unique identifier
Name	User name
Phone/Email	Contact detail
Preferred Language	Default language
Created At	Timestamp

Voice Session Entity

Field	Description
ID	Unique identifier

User ID	User reference
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Audio File	Voice input file
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Transcript	Converted text
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Confidence Score	STT confidence
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Created At	Timestamp
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Intent Entity

Field	Description
ID	Unique identifier
Session ID	Voice session reference
Intent Type	Place order / reorder / cancel / modify
Confidence Score	Prediction confidence
Created At	Timestamp

Order Entity

Field	Description
ID	Unique identifier
User ID	User reference
Restaurant ID	Restaurant reference
Items	Ordered items
Quantity	Number of items
Special Instructions	Notes from user
Status	Pending / confirmed / placed
Created At	Timestamp

Extracted Entity Entity

Field	Description
ID	Unique identifier
Order ID	Order reference
Item Name	Food item

Quantity	Quantity extracted
Restaurant Name	Restaurant reference
Language Type	English / Hindi / mixed
Created At	Timestamp

Assumptions and Dependencies

- The system assumes a working database connection
- The system depends on a backend framework for API handling
- The system depends on speech recognition and NLP libraries
- The system assumes valid audio input
- The system assumes user history is available for context-based ordering

Future Enhancements

- User authentication and roles
- Voice-based order editing
- Multilingual support for more Indian languages
- AI-based restaurant recommendations
- Payment integration
- Delivery tracking
- SMS or WhatsApp order confirmation
- Dashboard for analytics
- Integration with food delivery platforms

README File Should Include

The project **README** should clearly contain:

- Project overview
- Features
- Tech stack

- Installation steps
- Backend setup
- Frontend setup
- API endpoints
- Database setup
- How voice processing works
- Sample voice commands
- Mixed-language examples
- Folder structure
- Testing instructions
- Future scope

Approval

This SRS document serves as the baseline for design, development, and testing of the **Voice-to-Order AI System**.